

BAO VE Y TUONG AI

AirGuard-AI Pipeline

Vong so tuyen - Cuoc thi Thiet ke Vi mach Do thi Thong minh 2026

1. Tom tat

TinyML tren chip: MINA 1D-CNN, 4018 params, 6832 MACs, Q1.15 (0% drop). Co baseline, ablation, da seed, training on dinh (AdamW + scheduler + early stop).

2. Baseline doi chung (test)

Model	Acc	F1	Danger R
Logistic Reg	74.4%	75.0%	89.8%
Random Forest	76.6%	76.9%	86.2%
MLP	69.5%	69.6%	90.1%
MINA CNN	70.7%	71.3%	87.6%

RF accuracy cao hon nhung khong deploy duoc len chip. MINA: 4K params, Q1.15, safety layer dua Danger Recall len 99.7%.

3. Ablation study (test)

Config	Acc	F1	MACs
8ch W=24 CE	61.1%	60.7%	35520
+ transient	59.0%	58.3%	36192
+ focal	57.1%	57.0%	36192
+ W=3	69.6%	70.3%	6832
+ multi-task	70.3%	70.7%	6832
W=6 control	71.8%	72.0%	9768

W=3 la cai tien lon nhat: +10 diem macro-F1, MACs giam 5x.

4. Da seed (n=3)

Metric	Mean +/- Std
Accuracy	69.3% +/- 1.8%
Macro-F1	69.8% +/- 1.7%
Danger Recall	89.4% +/- 1.7%

5. Safety 3 lop

Mode	Danger R	FPR
argmax	87.6%	0.4%

asymmetric	99.7%	70.6%
fused OR	100%	91.5%

6. Lenh chay

- python src/baselines.py
- python src/ablation.py --epochs 12
- python src/run_seeds.py --seeds 42 1 7
- python src/train.py --scheduler cosine --patience 8

7. Q&A nhanh

Q: Tai sao CNN khong LSTM?

A: 4K params, map MINA HW, <10us inference.

Q: RF cao hon?

A: Trade-off deployability vs raw accuracy.

Q: Tai lap?

A: 3 seeds, std ~1.7%.

Q: Q1.15?

A: 0% accuracy drop verified.